PAPER_313: NGC 6302 Equatorial Torus Magnetic Confinement — UQFF Plasma β and Alfvén Analysis

Subtitle: FIRST UQFF Equatorial PN Torus Magnetic Confinement — η_B = 3.979×10^5; β_plasma = 2.513×10^{-6}; v_Alfvén = 8.921×10^7 m/s

Author: Daniel T. Murphy
Session: 89 | **Date:** March 17, 2026
Module: `NGC6302_UQFF_MODULE.cpp` (31st C++ UQFF module)
WOLFRAM_TERM: `NGC6302_TORUS_CONFINEMENT`
UQFF First: FIRST UQFF explicit equatorial torus magnetic confinement analysis (β_plasma < 10^{-5}, magnetically dominated regime)

Abstract

This paper presents UQFF derivations and numerical results for: PAPER_313: NGC 6302 Equatorial Torus Magnetic Confinement — UQFF Plasma β and Alfvén Analysis. Calibration constants: κ = 0.0005/day, [SSq] = 0.57. Results validated against observational data and prior UQFF whitepaper series.

1. System Parameters

Parameter	Value	Notes
B	1.0×10^{-5} T	Equatorial torus magnetic field
μ_0	1.2566×10^{-6} H/m	Permeability of free space
v_wind	1.0×10^5 m/s	Stellar wind (ram pressure driver)
ρ_fluid	1.0×10^{-20} kg/m^3	Ionized torus/lobe gas

---

2. Unique Physics

2.1 Magnetic Pressure in the Equatorial Torus

The equatorial dust and gas torus of NGC 6302 confines the bipolar outflow. The magnetic pressure in the torus:

$$P_{mag} = \frac{B^2}{2\mu_0} = \frac{(10^{-5})^2}{2 \times 1.2566 \times 10^{-6}}$$

$$= \frac{10^{-10}}{2.5133 \times 10^{-6}} = \mathbf{3.979 \times 10^{-5}} \text{ Pa}$$

2.2 Wind Ram Pressure

The ram pressure of the stellar wind acting on the torus:

$$P_{ram} = \rho \cdot v_{wind}^2 = 1.0 \times 10^{-20} \times (10^5)^2 = 1.0 \times 10^{-20} \times 10^{10} = \mathbf{1.0 \times 10^{-10}} \text{ Pa}$$

2.3 Magnetic Confinement Ratio

$$\eta_{B_{conf}} \equiv \frac{P_{mag}}{P_{ram}} = \frac{3.979 \times 10^{-5}}{1.0 \times 10^{-10}} = \mathbf{3.979 \times 10^5}$$

The equatorial torus magnetic pressure exceeds the stellar wind ram pressure by **3.979×10^5** — providing the confinement force that prevents the torus from being blown away by the wind and channels the outflow into two polar lobes.

2.4 Plasma Beta Parameter

$$\beta_{plasma} \equiv \frac{P_{ram}}{P_{mag}} = \frac{1.0 \times 10^{-10}}{3.979 \times 10^{-5}} = \mathbf{2.513 \times 10^{-6}}$$

$\beta_{plasma} \ll 1$ confirms the system is in the **magnetically dominated regime** (β « 1). Magnetic forces control the plasma dynamics; thermal and kinetic (ram) pressures are negligible compared to magnetic pressure.

2.5 Alfvén Velocity

The Alfvén velocity sets the characteristic speed of magnetic disturbance propagation in the torus:

$$v_A = \frac{B}{\sqrt{\mu_0 \rho}} = \frac{10^{-5}}{\sqrt{1.2566 \times 10^{-6} \times 10^{-20}}}$$

$$\sqrt{1.2566 \times 10^{-26}} = \sqrt{1.2566} \times 10^{-13} = 1.121 \times 10^{-13}$$

$$v_A = \frac{10^{-5}}{1.121 \times 10^{-13}} = \mathbf{8.921 \times 10^7} \text{ m/s}$$

2.6 Alfvén-to-Wind Velocity Ratio

$$\frac{v_A}{v_{wind}} = \frac{8.921 \times 10^7}{1.0 \times 10^5} = \mathbf{892.1}$$

The Alfvén velocity exceeds the stellar wind speed by a factor of **892.1**. This means magnetic signals propagate ~892× faster than the wind through the torus medium, enabling rapid magnetic restructuring and sustaining the stable, long-lived torus morphology observed in NGC 6302 over its ~2000 yr lifetime.

---

3. Physical Interpretation

The three coupled PAPER_313 quantities form a self-consistent magnetically dominated confinement picture:

Condition	Value	Interpretation
η_B_conf = P_mag/P_ram » 1	3.979×10^5	Magnetic confinement dominates

Condition	Value	Interpretation
β_plasma « 1	2.513×10^{-6}	Magnetically dominated plasma
v_A / v_wind » 1	892.1	Rapid magnetic signal propagation

Together these confirm: **the equatorial torus of NGC 6302 is a magnetically confined structure**. Magnetic pressure exceeds ram pressure by ~4×10^5, the plasma β is ~$10^6\times$ below unity, and the Alfvén velocity is ~9×10^7 m/s (~30% of c), all consistent with a stable, magnetically dominated toroidal barrier that channels bipolar outflow perpendicular to the torus plane.

---

4. UQFF Context

While P_mag is not directly included as an additive acceleration term in the UQFF pipeline (it acts as a confinement geometry parameter), its ratio η_B_conf modulates the superconductivity-analogue factor $(1 - B/B_{crit})$ and the torus stability that allows the bipolar geometry to exist. The torus acts as the boundary condition that forces all wind and radiation terms (PAPER_311, PAPER_312) to act preferentially along the polar axis.

---

5. Key Results

Quantity	Value	Unit
P_mag = $B^2/(2\mu_0)$	3.979×10^{-5}	Pa
P_ram = ρ v_wind^2	1.000×10^{-10}	Pa
η_B_conf	$\mathbf{3.979\times10^{5}}$	dimensionless
β_plasma	$\mathbf{2.513\times10^{-6}}$	dimensionless
v_Alfvén	$\mathbf{8.921\times10^{7}}$	m/s
v_A / v_wind	892.1	dimensionless
v_A / c	0.2974	(29.7% c)

---

6. Classification

- **UQFF First:** FIRST UQFF equatorial PN torus magnetic confinement ($\beta < 10^{-5}$, magnetically dominated)
- **Scale:** Stellar (PN torus/lobe scale, ~1 ly)
- **Physics category:** Magnetohydrodynamics / plasma β / Alfvén dynamics / magnetic confinement
- **Cross-references:** PAPER_311 (wind shock), PAPER_312 (UV radiation)